

Feedback-like algorithm implementation in chemostat*

Algorithm for the identification of the control triggers and their application to the bioreactor in chemostat.

Algorithm 1 Implementation of Feedback-Inspired Control in Chemostat

Input: Plant model, operational constraints, nominal initial state x^0 , time step Δt , switching tolerance ϵ

(A) Offline preanalysis

Initialize $T_1, T_3, T_4, T, C \leftarrow \text{zeros}(N + 1)$

for $i = 0, \dots, N$ **do**

$T(i) \leftarrow T_{\min} + \frac{i}{N} \cdot (T_{\max} - T_{\min})$

Solve OCP for horizon $T(i)$, returning:

$$t_1, t_3, t_4, x_3(t_2), x_7(t_2)$$

$$T_1(i) \leftarrow \frac{t_1}{T(i)}, T_3(i) \leftarrow \frac{t_3}{T(i)}, T_4(i) \leftarrow \frac{t_4}{T(i)}$$

$$C(i) \leftarrow \frac{\partial \mu}{\partial x_3}(x_3(t_2), x_7(t_2))$$

end for

$T'_3 \leftarrow T_3$ (where $T_3 \neq 0$), $T' \leftarrow T$ (where $T_3 \neq 0$)

Use *Simple Linear Regression* to fit the coefficients:

$$(a_1^*, b_1^*) \leftarrow \text{SLR}(T_1, T) \quad (a_c^*, b_c^*) \leftarrow \text{SLR}(C, T)$$

$$(a_3^*, b_3^*) \leftarrow \text{SLR}(T'_3, T') \quad (a_4^*, b_4^*) \leftarrow \text{SLR}(T_4, T)$$

Output (A): estimated $(a_1^*, b_1^*), (a_c^*, b_c^*), (a_3^*, b_3^*), (a_4^*, b_4^*)$

(B) Online observer design

Design observer obs such that:

$$\hat{x}(k) = \text{obs}(u(k), v(k), y(k))$$

where $\hat{x}(k) = (\hat{x}_1(k), \dots, \hat{x}_8(k))$

Output (B): online state estimates $\hat{x}(k)$

(C) Feedback controller execution

Initialize $\text{switched} \leftarrow \text{false}$, $k \leftarrow 0$, $u(0) \leftarrow u_{\max}$

Compute $\tau_1 \leftarrow a_1^* \cdot T_f + b_1^*$, $c \leftarrow a_c^* \cdot T_f + b_c^*$

Compute $\tau_3 \leftarrow a_3^* \cdot T_f + b_3^*$, $\tau_4 \leftarrow a_4^* \cdot T_f + b_4^*$

Measure initial output $y(0)$

$$\hat{x}(0) \leftarrow \text{obs}(u(0), v(0), y(0))$$

$$\hat{\psi}(0) \leftarrow \frac{\partial \mu}{\partial x_3}(\hat{x}_3(0), \hat{x}_7(0))$$

if $\tau_3 > 0$ **then**

$$v(0) \leftarrow v_{\max}$$

else

$$v(0) \leftarrow v_{\min}$$

end if

while $k < \lfloor T_f / \Delta t \rfloor$ **do**

if $k \leq \tau_1 \cdot T_f / \Delta t$ **then**

// Initial flash phase

$$u(k + 1) \leftarrow u_{\max}$$

else if $\text{switched} = \text{false}$ **then**

Set:

$$\text{cond}_1 \leftarrow \hat{\psi}(k) \leq c + \epsilon$$

Set:

$$\text{cond}_2 \leftarrow \hat{\psi}(k - 1) > c + \epsilon$$

if cond_1 and cond_2 **then**

$\text{switched} \leftarrow \text{true}$

$$u(k + 1) \leftarrow u_{\max}$$

else

$$u(k + 1) \leftarrow u_{\min}$$

end if

else

// After switch phase

$$u(k + 1) \leftarrow u_{\max}$$

end if

if $\tau_3 \cdot T_f \leq k \cdot \Delta t \leq \tau_4 \cdot T_f$ **then**

$$v(k + 1) \leftarrow v_{\min}$$

else

$$v(k + 1) \leftarrow v_{\max}$$

end if

Apply $u(k + 1)$ and $v(k + 1)$ to the plant

Measure $y(k + 1)$

$$\hat{x}(k + 1) \leftarrow \text{obs}(u(k + 1), v(k + 1), y(k + 1))$$

$$\hat{\psi}(k + 1) \leftarrow \frac{\partial \mu}{\partial x_3}(\hat{x}_3(k + 1), \hat{x}_7(k + 1))$$

$$k \leftarrow k + 1$$

end while

Output (C): control sequences $u(1), u(2), \dots, u(\lfloor T_f / \Delta t \rfloor)$ and $v(1), v(2), \dots, v(\lfloor T_f / \Delta t \rfloor)$

End Algorithm

Algorithm 2 $\text{SLR}(Y, X)$

Input: Y, X

$$n \leftarrow \dim(X), \hat{X} \leftarrow \text{mean}(X), \hat{Y} \leftarrow \text{mean}(Y)$$

$$a_n \leftarrow 0, a_d \leftarrow 0$$

for $i = 1, \dots, N$ **do**

$$a_n \leftarrow a_n + (X(i) - \hat{X})(Y(i) - \hat{Y})$$

$$a_d \leftarrow a_d + (X(i) - \hat{X})^2$$

end for

$$a \leftarrow \frac{a_n}{a_d}$$

$$b \leftarrow \hat{Y} - a\hat{X}$$

Output: a, b

End Algorithm

*B. Boerkmann, T. Bayen, M. Khammash and W. Djema, "From Bang-Bang Optimal Control to State-Triggered Optogenetic Feedback Strategies for Protein Production in Yeast," Preprint, 2026.